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A Confidence-Aware Adaptive Framework for Temporal Segmentation in Real-Time Bangla Sign Language Recognition

Research Proposal

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Research Field: Machine Learning

1. Research Problem

10 Bangla Sign Language (BDSL) is the most common mode of communication amongst the deaf and hearing-impaired people of Bangladesh. There has been a lot of progress made in recognizing through camera on this very specific task of recognizing the sign in a video which contains one sign alone. But natural sign language is not presented like that. It is presented in one stream, and the very first step is to determine the boundaries of each sign.

Temporal Segmentation is the choice. The mistakes that arise due to segmentation affect the classifier because any segment that ends the sign and begins to incorporate the following one will give an error despite its accuracy in functioning. There are not many cues from continuous signing because there are not many pauses, there could be either quick or slow transitions depending on signs and there could be individual rhythms and styles of signers. Segmentation is difficult.

Two straightforward solutions to the problem exist, but none are effective. The first solution consists of using a fixed algorithm, for instance, by declaring that there is a boundary whenever the speed of the hand drops below a particular threshold, which does not require any effort from the user, yet is highly ineffective for various signers. Full control by the signer (by either pushing a button or executing a gesture each time) is very effective, but it complicates the task of the signer.

2 This paper presents the design and analysis of such a segmentation algorithm for real-time interpretation of the BDSL. This process began with the testing of key press by hand, gesture to signify transition, and implicit velocity threshold; these three have become the benchmarking techniques, the latter becoming the fixed threshold method and key press by hand becoming the totally user-involved technique. The algorithm is a landmark-based algorithm and is much cheaper than the video-based deep learning methods [2].

2. Existing Literature and Research Gap

BDSL datasets and isolated recognition. Current BDSL resources are built around isolated signs. BdSLW60 offers 60 words in 9,307 samples from 18 signers [3]. MV-BSLD holds 4,000 videos of 50 words recorded from several camera views [4]. Sign3DNet applies a 3D CNN to Bengali word-level recognition [5]. The related MV-BSL resource reportedly includes a 107-sentence subset for continuous recognition [6], and Ban-Sign-Sent-9K-V1 lists 9,610 videos of 1,922 sentences [7]. These resources typically label a whole clip with one word, gloss, or sentence; the frame-level boundaries, transition intervals, and difficulty labels a segmenter needs are not part of that labelling. Figures come from public documentation and will be re-checked.

Sign segmentation in other sign languages. Renz et al. treat segmentation as a learning problem, first with temporal convolutional networks [1] and then with changepoint-modulated pseudo-labelling [8]. Ye et al. recognize American Sign Language gestures inside continuous video [9]. Other work targets sentence-like units [10] and individual signs in continuous sequences [11]. A 2026 study examines how segmentation models affect translation accuracy [12], which supports measuring recognition under each segmentation method rather than boundary accuracy alone.

Landmark-based real-time systems. MediaPipe Holistic and hand-landmark pipelines with recurrent models have been reported for real-time continuous recognition [2].

Gap. Three gaps follow. (1) BDSL lacks continuous sequences annotated with sign boundaries and transition intervals. (2) In the segmentation work reviewed here, the task is scored as detection, and user effort is not a measured quantity, so no result says what a given accuracy costs in intervention. (3) Fixed rules and learned detectors treat every boundary the same way, whether or not they are sure of it. This proposal targets all three.

3. Research Questions

Central question. *On signer-independent test data, does an adaptive policy that accepts, waits, or requests confirmation according to boundary confidence segment continuous BDSL more accurately than a fixed-threshold baseline, and how many confirmation requests does it need to do so?*

Hypothesis. On the frozen test signers, the adaptive policy raises boundary F1 by at least 5 points over the fixed-threshold baseline while issuing no more than 0.15 confirmation requests per true boundary. Both targets are set before any tuning. If they are missed, the results are reported as they stand.

Table 1. Supporting questions, evidence, and pre-set success criteria.

ID	Supporting question	Evidence	Success criterion
SQ1	Does the adaptive policy beat the fixed-threshold baseline on boundary detection, and does the gain reach sign recognition?	Boundary F1, mean absolute boundary error, temporal IoU; sign accuracy and macro-F1 under ground-truth, baseline, and adaptive segmentation	F1 gain ≥ 5 points; recognition closes at least half of the accuracy gap between baseline and ground-truth segmentation
SQ2	How does accuracy change as confirmation requests rise from none to every boundary?	Accuracy–intervention curve over the (τ_{low}, τ_{high}) grid; requests per true boundary	A point at ≤ 0.15 requests per boundary; curve above the straight line joining the baseline (0) and manual input (1.0)
SQ3	Which feature groups contribute most to reliable boundary confidence?	Drop-one-group ablation (change in boundary F1 and requests per boundary); permutation importance as a cross-check	A ranked list; a group counts as useful if removing it lowers F1 by more than the seed-to-seed spread
SQ4	Does boundary confidence track real difficulty?	Expected calibration error (ECE); share of requests that fall on boundaries labelled hard (Part C test split)	ECE ≤ 0.10; hard boundaries make up a larger share of requests than of all boundaries

4. Methodology

4.1 Pipeline and features

Video is captured at 30 fps and passed through MediaPipe [13], which returns 21 landmarks per hand plus body pose landmarks. The stages are: camera input, landmark extraction, feature computation, boundary model, boundary confidence, adaptive policy, segmented sign, and sign classifier. Gaps of up to three frames in hand landmarks are filled by linear interpolation; longer gaps set a hand-present flag to 0, which also exposes two-handed to one-handed transitions.

Let q_t be the palm centre (mean of hand landmarks 0, 5, 9, 13, 17), w_t the wrist, s_t the shoulder width (pose landmarks 11 and 12), m_t the mid-shoulder point, and n_t the nose. Dividing every distance by s_t removes the effect of how far the signer sits from the camera. Groups G1–G5 are computed for each hand.

Table 2. Frame-level feature groups (t = frame index; G6 and G7 are optional inputs).

Group	Definition
G1 Speed, acceleration	$v_t = fps \cdot \lvert q_t - q_{t-1} \rvert / s_t$; $a_t = fps \cdot (v_t - v_{t-1})$
G2 Motion stability	u_t = standard deviation of v over frames t−4 … t
G3 Direction change	θ_t = angle between δ_t and δ_{t-3}, where $\delta_t = q_t - q_{t-3}$
G4 Inter-hand distance	$d_t = \lvert q_t(\text{left}) - q_t(\text{right}) \rvert / s_t$
G5 Wrist to torso, face	$g_t = \lvert w_t - m_t \rvert / s_t$; $f_t = \lvert w_t - n_t \rvert / s_t$
G6 Pose movement	Mean per-frame displacement of shoulder and elbow landmarks, divided by s_t
G7 Classifier confidence	Maximum softmax probability of the sign classifier on the current window

4.2 Boundary model and boundary confidence

The model reads the last W = 15 frames (0.5 s) of features. A one-layer LSTM with 64 hidden units outputs a logit z_t for "frame t lies in a transition interval". Training uses weighted binary cross-entropy (transition frames are the minority class), the Adam optimizer, and early stopping on validation signers. Temperature scaling, fitted on validation signers only, gives a calibrated probability p_t = sigmoid (z_t / T). **Boundary confidence** is $C_t = |2p_t - 1|$: it equals 1 when the model is certain either way and 0 when $p_t = 0.5$. A decision point is raised at each local peak of p_t (±5 frames) above $p_{min} = 0.2$, and an accepted boundary is placed at the peak frame. The low p_{min} keeps recall high and leaves weak peaks to the adaptive policy.

4.3 Adaptive policy

Table 3. Adaptive policy at each decision point ($\tau_{low} \le \tau_{high}$).

Condition	Action	Behaviour
$C_t \ge \tau_{high}$	Accept	Commit the predicted label automatically: a boundary if $p_t \ge 0.5$, otherwise discard the candidate.
$\tau_{low} \le C_t < \tau_{high}$	Wait	Hold the decision and read up to $N_{wait} = 6$ more frames (200 ms), re-scoring each frame. Accept as soon as $C_t \ge \tau_{high}$; otherwise escalate to Request.
$C_t < \tau_{low}$	Request	Ask the signer to confirm (key press or open-palm gesture in the prototype).

The thresholds are chosen on validation signers by grid search (τ_{low} from 0.10 to 0.50, τ_{high} from τ_{low} + 0.10 to 0.90, step 0.05), maximizing boundary F1 subject to at most 0.15 requests per true boundary. Offline, a Request is answered from the ground-truth labels; a second run flips 10% of answers to test sensitivity to user error. Wait adds at most 200 ms of latency.

4.4 Dataset: BDSL-ATB

BDSL-ATB (Bangla Sign Language Adaptive Temporal Boundary dataset) is a specialised benchmark layer, not a replacement for large BDSL corpora. Its value lies in annotations that existing datasets lack: each sign stored as an interval $B_i = [f_{start}, f_{end}]$ (transitions are gradual, so a single frame would overstate precision), labelled transition intervals, boundary-difficulty labels, and the expected action at each boundary. It has three parts.

- **Part A, public isolated-sign data.** BdSLW60 [4] and MV-BSLD [5] train and validate the sign classifier and reduce new collection. Ban-Sign-Sent-9K-V1 [8] is a candidate for sentence-level exploration once four points are verified: reliable download, per-sign timing in glosses, signer IDs, and license terms. All landmarks are re-extracted with one MediaPipe version at 30 fps so features match across sources.
- **Part B, new continuous sequences.** 15–20 signers, 20 signs drawn from the BdSLW60 vocabulary, 10 sequence templates of 3–4 signs, and 5 repetitions per signer per template: 750 recordings and 2,250–3,000 sign instances at 15 signers (1,000 recordings at 20). A fluent BDSL signer or expert validates every template, because a Bangla sentence does not necessarily map to a valid BDSL sign sequence.

- **Part C, adaptive-interaction test set.** Deliberately hard transitions: fast signing with no artificial pause, similar hand positions, direction changes, two- to one-handed transitions, hesitations, and mild occlusion, recorded across signers.

Annotation. Sign intervals, transition intervals, and difficulty are marked in a frame-indexed spreadsheet template. A boundary is labelled *hard* when three or more of the seven Part C criteria hold, *medium* when two hold, and *easy* otherwise, mapping to the expected Request, Wait, and Accept. A second annotator labels 20% of sequences; agreement is reported as mean boundary offset in frames and Cohen's κ on frame-level labels. Participants give written informed consent, and video is released only where a signer agrees.

Signer-independent split. 14 training, 3 validation, and 3 test signers for a pool of 20 (10/2/3 if only 15 are recruited), in line with the signer-aware split reported for BdSLW60. No signer appears in two partitions, so the test measures unseen signing styles, not just held-out recordings. The test set is frozen before τ_{low} and τ_{high} are tuned. Part C follows the same partition; only its test portion is used for SQ4.

4.5 Baselines and evaluation

Baselines. B1 is a fixed-threshold segmenter: a boundary is declared when v_t stays below one global threshold for k consecutive frames, with the threshold and k tuned once on training signers and applied unchanged to every signer. B2 is manual key-press segmentation, simulated with ground-truth boundaries at one request per boundary. The transition-gesture condition costs one action per boundary plus a dedicated gesture, so it is treated as a variant of B2.

Segmentation. A predicted boundary is a true positive if it falls within 5 frames (about 167 ms) of an annotated transition interval, and each interval matches at most one prediction; 3 and 8 frames serve as sensitivity checks. Metrics are precision, recall, and F1; mean absolute boundary error against the interval midpoint, in milliseconds; and temporal IoU between predicted and true sign segments. F1 is reported after simulated confirmation, next to the precision of Accept decisions alone.

Recognition. The classifier is trained once on Part A and the Part B training signers. Sign accuracy and macro-F1 are computed on segments from ground-truth, B1, and adaptive segmentation, showing how much of the gap to ground-truth segmentation the adaptive policy closes.

Real-time and interaction. Frames per second (target ≥ 25 on a laptop CPU, no GPU), end-to-end latency (mean and 95th percentile), per-frame processing time, and dropped-frame rate. Interaction is measured by requests per true boundary, the autonomy rate (share of boundaries resolved without a request), and the accuracy–intervention curve with B1 at 0 and B2 at 1.0 as reference points.

Statistics. Differences carry 95% bootstrap confidence intervals resampled at signer level; every learned model is trained with three random seeds.

4.6 Ablation design

Table 4. Ablation variants. Each reports boundary F1, requests per true boundary, and ECE.

Variant	Change	Question answered
A1 No Wait	Wait removed; a single threshold splits Accept from Request	Does deferring the decision help?
A2 Fixed cut-off	Learned model with $p_t \geq 0.5$; no confidence, no requests	Does confidence add value beyond the learned model?
A3 Uncalibrated	Raw sigmoid(z_t) with no temperature scaling	Does calibration matter?
A4–A9 Drop one group	Retrain without G1, G2, G3, G4, G5, or G6, one at a time	Which features carry the confidence estimate? (SQ3)
A10 Add G7	Add sign-classifier confidence as an input	Does recognition feedback improve segmentation?

4.7 Tools and technology stack

Python with OpenCV (capture), MediaPipe (landmarks), NumPy and Pandas (time series), scikit-learn (baselines, splits, metrics), and PyTorch or TensorFlow/Keras (LSTM boundary model and lightweight sign classifier). Matplotlib and Seaborn produce figures; Jupyter and GitHub handle experiments and version control. A Streamlit or OpenCV interface shows camera input, current sign, boundary confidence, and confirmation prompts live.

4.8 Phased development plan

Table 5. Phases and gates. Week ranges are the proposed schedule.

Phase	Work	Output and gate
P1 Weeks 1–3	Record pilot sequences with a few signers; draft the annotation protocol; second-annotator check	Documented protocol. Gate: $\kappa \geq 0.80$ and mean boundary offset ≤ 3 frames before scaling up
P2 Weeks 4–8	Build B1, features, boundary model, confidence, and policy; small ablation; webcam demo	Pilot results (the original two-month scope); enough for a proof-of-concept paper
P3 Weeks 9–14	Collect Parts B and C; integrate Part A; freeze the signer-independent split	Frozen test set before any threshold tuning
P4 Weeks 15–20	Tune thresholds on validation signers; evaluate on test; full ablation; real-time benchmarks; write-up	Complete empirical study, or an extended pilot if P3 is only partly done

5. Research Topic: Innovation and Urgency

Innovation: Segmentation work usually asks one question: where is the boundary? This proposal adds two more: how sure is the system, and what should it do about it? Boundary confidence becomes a calibrated, tested quantity that decides between automation and user input, and user effort becomes a reported metric alongside F1. In the literature reviewed here, neither appears for BDSL. BDSL-ATB adds frame-level sign and transition labels with difficulty metadata, and its value is that annotation, not its size.

Urgency: BDSL has isolated-word resources [3, 4] and growing sentence-level data [7], yet recognizers trained on them cannot handle live signing because nothing reliably cuts the stream into signs. Temporal segmentation is the missing link between isolated sign classification and real-time continuous BDSL interpretation. Without it, better classifiers and bigger datasets do not become a usable tool. The landmark-based design needs only an ordinary webcam and no GPU.

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